Tutorial 3: Linear Transformations

Question Set (Practice Values)

INF1004 Mathematics II

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Question 1

Cramer's Rule

Determine the values of i_1, i_2, and i_3 for the following system of linear equations using **Cramer's rule**:

$$\begin{aligned} i_1 + i_2 + i_3 &= 6 \\ i_1 - i_2 + i_3 &= 0 \\ 2i_1 + i_2 - i_3 &= 6 \end{aligned}$$

Question 2

Gaussian Elimination

Use **Gaussian elimination** to solve the following system of three equations:

$$
\begin{aligned}
a + b + c &= 2 \\
2a + 2b + 3c &= 6 \\
3a - b - c &= 2
\end{aligned}
$$

Question 3

Consistency and Inverse Matrix Method

Solve the following systems of linear equations using Gaussian Elimination. Identify which system is **inconsistent**:

(a)

$$x + y = 3$$
$$2x - y = 0$$

(b)

$$x - 2y = 4$$
$$2x - 4y = 10$$

(c)

$$u + v + w = 5$$
$$2u + 2v + 2w = 10$$
$$u - v + w = 0$$

Question 4: Determining Inverses

We determine whether the matrices A, B, C are invertible and compute the inverses when they exist.

Matrix A:

$$A = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 2 \end{pmatrix}$$

Matrix B:

$$B = \begin{pmatrix} 1 & 1 & 1 \\ 2 & 2 & 2 \\ 0 & 1 & 3 \end{pmatrix}.$$

Matrix C:

$$C = \begin{pmatrix} 2 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Question 5

Real-world Application (MMORPG)

In an MMORPG, you can equip three types of items: **Mythril Helm (x)**, **Dragon Plate (y)**, and **Speed Boots (z)**. The stats for each are:

Item	**Attack**	**Defense**	**Agility**
Mythril Helm (x)	5	15	2
Dragon Plate (y)	20	25	0
Speed Boots (z)	2	5	10

To clear the boss, you need to reach a **Total Attack of 235** and a **Total Defense of 350**. If the **total number of items** you can equip is exactly **20**, determine how many of each item you should equip.

Hint: Form a system of linear equations.

Question 6

3D Rotations

Given Information: In 3D space in computer graphics, rotation matrices that rotate non-zero vectors at angles θ (about the x-axis) and ϕ (about the y-axis) are defined as:

$$R_x(\theta) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{pmatrix} \quad \text{and} \quad R_y(\phi) = \begin{pmatrix} \cos\phi & 0 & \sin\phi \\ 0 & 1 & 0 \\ -\sin\phi & 0 & \cos\phi \end{pmatrix}$$

Questions:

1. **Individual Matrices:** Calculate $A = R_x(90^\circ)$ and $B = R_y(90^\circ)$.
2. **Order 1: x-axis then y-axis:** Calculate the combined rotation matrix M_1 that represents rotating first around the x-axis, and then around the y-axis. $M_1 = B \cdot A = R_y(90^\circ)R_x(90^\circ)$ *(Note: The matrix on the right acts first on the vector.)*
3. **Order 2: y-axis then x-axis:** Calculate the combined rotation matrix M_2 that represents rotating first around the y-axis, and then around the x-axis. $M_2 = A \cdot B = R_x(90^\circ)R_y(90^\circ)$. Compare the matrices M_1 and M_2. Are they equal? Briefly explain what this implies about the sequence of 3D rotations.